

ECR-2 SIMULATOR

LLX-RND-2026-0003 | Numerical Run Report

Run #711 23 Aug 2026, 10:24:52 PM IST

Preliminary counter-current extraction prediction

This report reproduces the persisted Run #711 result snapshot. It is an audit of the completed preliminary Simulator calculation at the user-selected 50 deg C; it is not a release-grade rating, fabrication basis, or final engineering validation.

RESULT: BVP CONVERGED AND BOTH COMPONENT/GLOBAL BALANCES PASSED. THERMODYNAMIC APPLICABILITY AT 50 deg C IS PRELIMINARY / EXTRAPOLATED UNDER TASK #124. RELEASE STATUS: NOT RELEASE ELIGIBLE.

Run identity

Item	Value
Run ID	#711
Revision	8
Persisted database time	Sun Aug 23 2026 16:54:52 GMT+0000 (Coordinated Universal Time)
User-facing time	23/08/2026, 10:24:52 pm IST
Selected operating temperature	50 deg C / 323.15 K
Phase configuration	NMP continuous / RRBO dispersed
Calculation status	warning
Simulator status	CALCULATED_PRELIMINARY
Release status	NOT_RELEASE_ELIGIBLE

Headline numerical results

Item	Value
RRBO recovery to raffinate	81.9882%
Raffinate outlet	2895.25554 kg/h
Extract outlet	4556.62446 kg/h
Hydrocarbon-only raffinate aromatic wt%	20.93 wt%
Saturates recovery	86.4373%
Total aromatic removal	31.3592%
Active extraction height	0.75 m
Numerical compartments	3

1. Process basis and geometry

Item	Value
RRBO feed	3448 kg/h
NMP solvent feed	4024 kg/h
Solvent-to-oil mass ratio	1.167053
Column diameter	0.85 m
Rotor diameter	0.425 m
Compartment height	0.25 m
Rotor-to-column ratio	0.5
RRBO volumetric flow	4 m3/h
NMP volumetric flow	4 m3/h
Specific throughput	14.09815 m3/m2/h

2. Outlet flows and component performance

Component	Feed kg/h	Extract kg/h	Raffinate kg/h	Extract / removal	Raffinate recovery
Sat	2586.00259	350.73146	2235.27113	13.5627%	86.4373%
Mono	287.33247	102.3334	184.99907	35.615%	64.385%
Di	287.33247	94.65315	192.67932	32.942%	67.058%
Poly	287.33247	73.32901	214.00346	25.5206%	74.4794%
NMP	4003.88	3935.57744	68.30256	98.2941%	1.7059%
Total	7451.88	4556.62446	2895.25554	-	-

Hydrocarbon-only raffinate aromatic mass fraction: 20.93 wt%. NMP is excluded from the product denominator.

Sulfur / DBT prediction: NOT_IMPLEMENTED. Aromatic-transfer results must not be interpreted as sulfur or DBT removal.

3. Component and global mass balances

Component	Feed	Extract	Raffinate	Residual kg/h	Status
Sat	2586.002586	350.7314592	2235.2711268	-6.7473138e-11	PASS
Mono	287.3324713	102.3334003	184.999071	5.1159077e-13	PASS
Di	287.3324713	94.6531498	192.6793215	1.4210855e-13	PASS
Poly	287.3324713	73.3290076	214.0034638	7.1054274e-14	PASS
NMP	4003.88	3935.5774399	68.3025601	-6.6575012e-10	PASS
Global total	7451.88	4556.6244568	2895.2555432	-7.3249851e-10	PASS

Acceptance limits: total balance <= 5.0e-6 kg/h; maximum component balance <= 1.0e-6 kg/h.

4. Complete axial compartment profile

Component order for all arrays: [Sat, Mono, Di, Poly, NMP]. The stored axial state reports continuous/extract-side and dispersed/raffinate-side state totals at each compartment centre.

Comp.	z centre m	z interval m	Extract-side state kg/h	Raffinate-side state kg/h	phi_d	Re_d	a m2/m3
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Comp.	z centre m	z interval m	Extract-side state kg/h	Raffinate-side state kg/h	phi_d	Re_d	a m2/m3
1	0.125	0 - 0.25	4477.04787	3368.42341	0.210485	9.395261	1282.03028
2	0.375	0.25 - 0.5	4307.6709	3199.04645	0.200526	9.263795	1221.37007
3	0.625	0.5 - 0.75	4110.87526	3002.2508	0.189115	9.105749	1151.868

Comp.	d32 mm	U slip m/s	Raffinate mass fractions...	Extract mass fractions...
1	0.9850881	0.0118473	[0.754276, 0.079875, 0.080092, 0.081531, 0.004224]	[0.068226, 0.018775, 0.017222, 0.013542, 0.882235]
2	0.9850881	0.01168152	[0.762576, 0.073048, 0.073928, 0.078058, 0.012391]	[0.047414, 0.011302, 0.010173, 0.008289, 0.922822]
3	0.9850881	0.01148223	[0.769233, 0.066881, 0.068856, 0.075192, 0.019838]	[0.01804, 0.003842, 0.003416, 0.002856, 0.971846]

5. Axial hydrodynamics and transfer coefficients

All transfer arrays below use [Sat, Mono, Di, Poly, NMP]. Values are local preliminary physics outputs; governed release values remain unavailable.

Compartment	Metric	Sat	Mono	Di	Poly	NMP
1	Shc	38.666022	39.642371	40.737157	42.18433	30.696159
	Shd	34.055129	34.506275	35.009172	35.669066	29.825448
	kc m/s	1.161361e-5	1.113990e-5	1.064920e-5	1.005871e-5	1.739492e-5
	kd m/s	6.257783e-7	5.932258e-7	5.598975e-7	5.203353e-7	1.130116e-6
	Koverall m/s	5.356513e-7	5.827581e-7	5.573525e-7	5.100003e-7	1.126358e-6
	Koa s-1	0.0006867	0.0007471	0.0007145	0.0006538	0.001444

Holdup phi = 0.2104855; interfacial area = 1282.0302759 m2/m3; Re_d = 9.3952614.

Compartment	Metric	Sat	Mono	Di	Poly	NMP
2	Shc	38.830742	39.809364	40.906621	42.356936	30.839708
	Shd	33.774545	34.218506	34.713408	35.362835	29.612947
	kc m/s	1.166308e-5	1.118682e-5	1.069350e-5	1.009987e-5	1.747627e-5
	kd m/s	6.206225e-7	5.882785e-7	5.551674e-7	5.158681e-7	1.122064e-6
	Koverall m/s	5.270954e-7	5.788049e-7	5.529011e-7	5.059132e-7	1.118901e-6
	Koa s-1	0.0006438	0.0007069	0.0006753	0.0006179	0.0013666

Holdup phi = 0.2005262; interfacial area = 1221.3700713 m2/m3; Re_d = 9.2637946.

Compartment	Metric	Sat	Mono	Di	Poly	NMP
3	Shc	38.989534	39.969861	41.068935	42.521499	30.981399
	Shd	33.437979	33.87331	34.35861	34.995464	29.358144
	kc m/s	1.171078e-5	1.123193e-5	1.073593e-5	1.013911e-5	1.755656e-5
	kd m/s	6.144379e-7	5.823439e-7	5.494932e-7	5.105089e-7	1.112409e-6
	Koverall m/s	5.181586e-7	5.737745e-7	5.474735e-7	5.010081e-7	1.109726e-6
	Koa s-1	0.0005969	0.0006609	0.0006306	0.0005771	0.0012783

Holdup phi = 0.1891152; interfacial area = 1151.8680036 m2/m3; Re_d = 9.105749.

6. Driving forces and component transfer rates

Positive transfer is RRBO/dispersed phase to NMP/continuous phase. Negative NMP values indicate transfer in the opposite direction.

Compartment	Metric	Sat	Mono	Di	Poly	NMP
1	Driving force kg/m3	258.218248	95.8084503	96.1730308	76.0662263	-38.5894406
	Rate kg/m3/s	0.17732393	0.07157979	0.06871975	0.04973482	-0.05572413
	Amount kg/h	90.5602452	36.5561678	35.0955305	25.3998304	-28.4585985
Compartment	Metric	Sat	Mono	Di	Poly	NMP
2	Driving force kg/m3	340.2082112	94.6925733	91.2570786	77.4644308	-32.035586
	Rate kg/m3/s	0.21901876	0.06694149	0.06162562	0.04786584	-0.04377959
	Amount kg/h	111.8540077	34.1873639	31.472521	24.4453298	-22.3584634
Compartment	Metric	Sat	Mono	Di	Poly	NMP
3	Driving force kg/m3	486.5817891	93.5908982	87.2046928	79.6804055	-26.7848647
	Rate kg/m3/s	0.29041651	0.06185539	0.05499278	0.04598318	-0.03423795
	Amount kg/h	148.3172064	31.5898687	28.0850982	23.4838474	-17.4854982

7. d32, holdup, power and property basis

Item	Value
d32 nominal	0.9850881 mm
d32 mode	direct_turbulence_preliminary
d32 equation	$d32 = C \times (\gamma / \rho_c)^{0.6} \times \epsilon^{-0.4}$
d32 C nominal / sensitivity	0.39 / 0.36 - 0.43
d32 sensitivity	0.909 - 1.086 mm
d32 status	DIRECT_TURBULENCE_D32_PRELIMINARY; not pilot validated
Global holdup phi	0.2050035
Global interfacial area	1248.6407723 m2/m3
Interfacial tension	0.01 N/m
Sigma basis	Governed preliminary constant-over-range basis; sigma(T) is not modelled
Sigma anchor	0.01 N/m at 70 deg C, retained unchanged
Power density	98.3274756 W/m3
Specific power psi	0.097741 W/kg
Rotor speed	60 rpm
Continuous density basis	1006 kg/m3
Dispersed density basis	862 kg/m3

8. Solver convergence and acceptance

Item	Value
BVP status	converged
Convergence status	accepted
Iterations	13
Function evaluations	404
Final residual norm	1.044585361e-13
Maximum normalized residual	7.643532317e-14
Relative state change	7.876397009e-11
Total balance residual	-7.324985063e-10 kg/h
Maximum component balance residual	3.060378617e-10 kg/h
Property validity status	passed
BVP acceptance termination	accepted
Transfer status	LOCAL_PRELIMINARY_CALCULATED
Governed transfer values	UNAVAILABLE
Release status	NOT_RELEASE_ELIGIBLE

9.Task #124 thermodynamic temperature status

THERMODYNAMIC STATUS: PRELIMINARY / EXTRAPOLATED AT 50 deg C. This is a temperature-applicability classification, not an additional execution gate.

Item	Value
Temperature status	EXTRAPOLATED
Selected temperature	50 deg C / 323.15 K
Nearest admitted evidence	298.15 K
Distance outside admitted range	25 K
Selected-temperature applicability	EXTRAPOLATED_PRELIMINARY
NRTL model	LLX_TLLE_NRTL_TAU_B_OVER_T
Parameter identifiability	CALIBRATION_TEMPERATURE_ONLY
Independent multi-temperature validation	INSUFFICIENT_EVIDENCE
Optimizer-forward readiness	NOT_READY
Calibration reproduction	FAIL
Tie-lines within gate	2/13
Maximum calibration deviation	0.092

Task #124 warnings:

- The active NRTL parameters are regressed at 298.15 K only; $\tau(T)=b/T$ is not independently temperature-validated.
- Coto calibration-temperature NRTL reproduction: FAIL (2/13 tie-lines within 0.009000000000000001; maximum $|\delta\tau| = 0.092$).

- Temperature Extrapolation - Preliminary / Pending Validation

10. Persisted run warnings and limitations

- INPUT_WARNING
- INTERFACIAL_TENSION_TEMPERATURE_DEPENDENCE_NOT_MODELLED
- PHYSICAL_MW_ASSUMED
- NMP_PURITY_PHYSICAL_ONLY
- EPD_ASSUMED_VALUE (x6)
- D32_DIRECT_TURBULENCE_PRELIMINARY
- INTERFACIAL_AREA_COMPUTED
- The sigma temperature-dependence warning is intentional: the unchanged 70 deg C anchor is held constant over the bounded preliminary range; no sigma(T) correlation was applied.
- The direct-turbulence d32 route is separate from K&H 1996 and is not pilot validated.
- The dispersed Re_d profile is approximately 9.11 - 9.40, below the reported rigid-sphere validity floor of 10.
- K&H-derived hydrodynamic and mass-transfer outputs are local preliminary physics, not governed release coefficients.
- Sulfur and DBT prediction remains NOT_IMPLEMENTED.

Audit conclusion

The persisted Run #711 is numerically coherent: the counter-current BVP converged, all component balances passed, and the transfer directions are internally consistent. The prediction remains PRELIMINARY / EXTRAPOLATED at 50 deg C and is not release eligible.

This report is an audit of the saved Simulator result only. It does not tune or modify the model, inputs, NRTL, BVP, K&H, physical-property routes, d32, holdup, C2, partition governance, or validation gates.